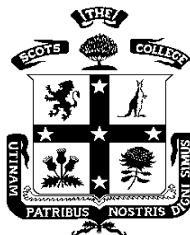

THE SCOTS COLLEGE



MATHEMATICS EXTENSION I

YEAR 12 ASSESSMENT TASK 3

4TH JUNE 2012

GENERAL INSTRUCTIONS

- Working time – 45 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided
- Show all necessary working
- *Each question must be completed in a new answer booklet.*
- Label each answer booklet with your name, class teacher's name and the question number attempted

WEIGHTING

20%

TOTAL MARKS

32

- a) Use the principle of mathematical induction to show that $8^n - 7n + 6$ is divisible by 7 for all positive integral values of n . **4 marks**
- b) A certain rectangle, with its length double its width, is also a cyclic quadrilateral.
- i. Let the length of the rectangle be y . **1 mark**
- Draw a diagram of the given information.
- ii. Show that the radius of the circle, r , in terms of y is $\frac{y}{4}\sqrt{5}$ **2 marks**
- iii. If the length of the rectangle, y , is decreasing at a rate of 3cm per second, calculate the rate the area of the circle is changing when the length of the rectangle is 4cm **3 marks**

- a) A particle in simple harmonic motion starts from a displacement of -6 units and takes eight seconds to travel to a displacement of 2 units. The particle oscillates between -6 units and 2 units.
- i. State the period of the particle **1 mark**
- ii. Sketch the displacement–time graph for $0 \leq t \leq 24$ **3 marks**
- iii. State the amplitude and centre of motion of the particle **2 marks**
- iv. Write down an equation for the position of the particle x at time t seconds in the form $x = a \cos(nt) + b$ where a , b and n are constants **2 marks**
- v. Hence, prove the particle moves in simple harmonic motion **2 marks**

- b) The motion of a particle is described by the acceleration equation

$$\frac{d^2x}{dt^2} = x$$

where x is the displacement of the particle from the origin in metres and t is time in seconds

- i. Initially the particle is at the origin travelling with a velocity of -3m/s . **3 marks**

Show that the velocity of the particle, in terms of x , is: $v = -\sqrt{x^2 + 9}$

- ii. By using the Standard Integrals Table, evaluate $\frac{d}{dx} \ln(x + \sqrt{x^2 + 9})$ **1 mark**

- iii. Hence, or otherwise, find a time when the particle is 4 metres to the left of the origin. Give your answer to the nearest second. **3 marks**

QUESTION 3 – ANSWER IN A NEW BOOKLET

6 MARKS

- a) Morgan deposits \$20,000 in a savings account with 6% per annum interest calculated at the end of each month.

- i. Write an expression for the value of Morgan's deposit at the end of the third month. **1 mark**

- ii. From the fourth month Morgan decides to withdraw \$800 just after interest is calculated and he continues to withdraw \$800 every month thereafter. **3 marks**

Let B_n be the balance of Morgan's savings account at the end of the n^{th} \$800 withdrawal.

Show that $B_n = 20000(1.005)^{n+3} - 160000(1.005^n - 1)$

- iii. Hence, how many times can Morgan withdraw \$800 from his account? **2 marks**

END OF EXAM

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Q1) a) ① Prove true for $n=1$

$$8 - 7 + 6 = 7$$

Divisible by 7

$\therefore$ true for $n=1$

② Assume true for $n=k$:

$$8^k - 7k + 6 = 7M \quad \text{where } M \text{ is an integer.}$$

③ Prove true for $n=k+1$

$$8^{k+1} - 7(k+1) + 6$$

$$(8^k)8 - 7k - 7 + 6 \quad 8^k = 7M + 7k - 6$$

$$(8^k)8 - 7k - 1$$

$$(7M + 7k - 6)8 - 7k - 1$$

$$56M + 56k - 48 - 7k - 1$$

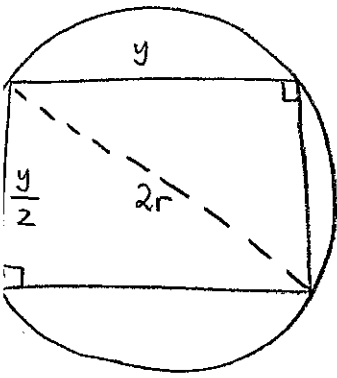
$$56M + 49k - 49$$

$$7(8M + 7k - 7)$$

$\therefore$ Divisible by 7

True for $n=k+1$

④ Since true for $n=1$ and $n=k+1$, then true for $n=2$. Since true for $n=2$ and $n=k+1$ then true for $n=3$ and so forth for all positive values of n .



$$\text{ii) } y^2 + \left(\frac{y}{2}\right)^2 = (2r)^2$$

$$y^2 + \frac{y^2}{4} = 4r^2$$

$$\frac{5y^2}{4} = 4r^2$$

$$r = \sqrt{\frac{5y^2}{16}} = \frac{y}{4}\sqrt{5}$$

$$= -3 \text{ cm/s}$$

$$= \frac{dy}{dt} \times \frac{dA}{dy}$$

$$= -3 \times \frac{5\pi y}{8}$$

$$y = 4.$$

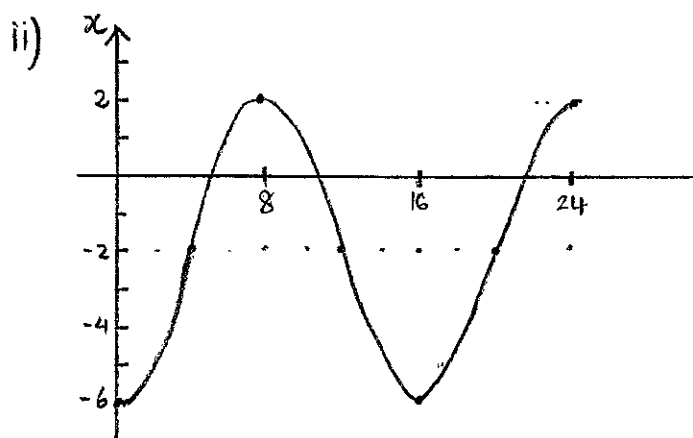
$$= -\frac{15\pi}{2} \text{ cm}^2/\text{s}$$

$$A = \pi r^2$$

$$A = \pi \left(\frac{5y^2}{16}\right)$$

$$\frac{dA}{dy} = \frac{5\pi y}{8}$$

Q2) a) i) Period = 16 seconds.



iii) Amplitude = 4
Centre of motion = -2

iv) $x = a \cos(nt) + b$

$$x = 4 \cos(nt) - 2$$

$$16 = \frac{2\pi}{n} \quad \therefore n = \frac{2\pi}{16} = \frac{\pi}{8}$$

$$\therefore x = 4 \cos\left(\frac{\pi}{8}t\right) - 2$$

v) $v = -\frac{4\pi}{8} \sin\left(\frac{\pi}{8}t\right)$

$$= -\frac{\pi}{2} \sin\left(\frac{\pi}{8}t\right)$$

$$a = -\frac{\pi^2}{16} \cos\left(\frac{\pi}{8}t\right)$$

Note: $\frac{x+2}{4} = \cos\left(\frac{\pi}{8}t\right)$

$$a = -\frac{\pi^2}{64} (x+2)$$

This is in the form $a = -n^2(x+A)$
 $\therefore$ SHM.

Q2) b) $a = z$

i) $t=0, x=0, v=-3$

$$\frac{d}{dx} \frac{1}{2} v^2 = z$$

$$\frac{1}{2} v^2 = \frac{x^2}{2} + C_1$$

$$\frac{1}{2} (9) = 0 + C_1$$

$$C_1 = \frac{9}{2}$$

$$\frac{1}{2} v^2 = \frac{x^2}{2} + \frac{9}{2}$$

$$v^2 = x^2 + 9$$

$$v = \pm \sqrt{x^2 + 9}$$

When $x=0, v=-3$

$$\therefore v = -\sqrt{x^2 + 9}$$

ii) $\frac{d}{dz} \ln(x + \sqrt{x^2 + 9})$

$$= \frac{1}{\sqrt{x^2 + 3^2}} = \frac{1}{\sqrt{x^2 + 9}}$$

iii) $t = ?$ when $x = -4$

$$v = -\sqrt{x^2 + 9}$$

$$\frac{dx}{dt} = -\sqrt{x^2 + 9} \quad \therefore dt = \int \frac{-1}{\sqrt{x^2 + 9}} dz$$

$$t = \int \frac{-1}{\sqrt{x^2 + 9}} dx$$

$$t = -\ln(x + \sqrt{x^2 + 9}) + C_2$$

$$0 = -\ln(0 + \sqrt{9}) + C_2$$

$$C_2 = \ln(3)$$

$$\therefore t = -\ln(x + \sqrt{x^2 + 9}) + \ln(3)$$

When $x = -4, t = ?$

$$t = -\ln(-4 + \sqrt{16 + 9}) + \ln 3$$

$$= -\ln(1) + \ln 3$$

$$t = \ln 3$$

$$\sim 1 \text{ second}$$

Q3 a) i) $20000(1.005)^3$

ii) $B_1 = 20000(1.005)^4 - 800$

$B_2 = 20000(1.005)^5 - 800(1.005) - 800$

$B_3 = 20000(1.005)^6 - 800(1.005)^2 - 800(1.005) - 800$

$B_4 = 20000(1.005)^7 - 800(1.005)^3 - 800(1.005)^2 - 800(1.005) - 800$
 $= 20000(1.005)^7 - 800[1.005^3 + 1.005^2 + 1.005 + 1]$

$\therefore B_n = 20000(1.005)^{n+3} - 800[1.005^{n-1} + 1.005^{n-2} + \dots + 1.005 + 1]$
 $= 20000(1.005)^{n+3} - 800 \left[\frac{1(1.005^n - 1)}{1.005 - 1} \right]$

i) $B_n = 20000(1.005)^{n+3} - 160000(1.005^n - 1)$

ii) $B_n = 20000(1.005)^n(1.005)^3 - 160000(1.005)^n + 160000 = 0$
 $= 4000[(1.005)^n(5(1.005)^3 - 40) + 40] = 0$

$(1.005)^n = \frac{-40}{5(1.005)^3 - 40}$

$n = \ln\left(\frac{-40}{5(1.005)^3 - 40}\right) \div \ln(1.005)$
 $= 27.205$

$\therefore$ Morgan can withdraw \$800 27 times.